Partial Differential Equations

I. Wave equation

1. Definition, physical derivation and classification.

Def I.1.1. Let $u = u(x,t)$ be a real-valued function on $\mathbb{R}^d \times \mathbb{R}$, where t is interpreted as time.

Let $\Delta = \frac{\partial^2}{\partial x_1^2} + \dots + \frac{\partial^2}{\partial x_n^2}$ be the Laplacian, $c > 0$ be a constant (wave speed)

The wave equation is $u_{tt} - c^2 \Delta u = 0$. Notation: $\Box := \frac{\partial^2}{\partial t^2} - c^2 \Delta$ is called the wave operator.

Rmk. 1. If u, v and solutions, so is $\alpha u + \beta v \quad \forall \alpha, \beta \in \mathbb{R}$.

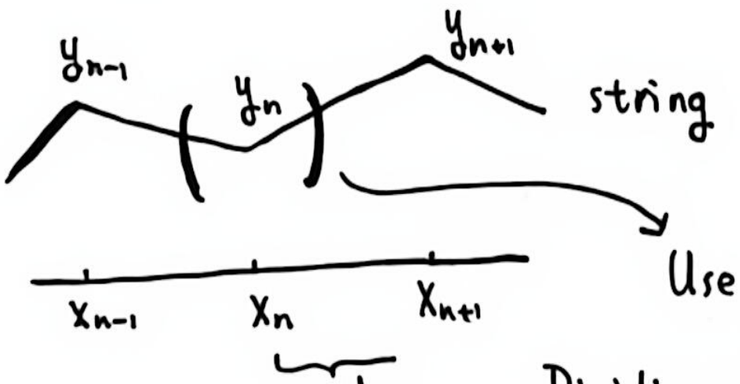
2. The equation is translation-invariant and invariant under the scaling $u(x,t) \mapsto u(\lambda x, \lambda t)$ λ:

3. The equation is invariant under rotations of $\mathbb{R}^d$.

4. the Cauchy problem: Given f, g on $C^\infty(\mathbb{R}^d)$, find u satisfying

$$\begin{cases} \frac{\partial^2}{\partial t^2} u - c^2 \Delta u = 0 \\ u(x,0) = f(x) \rightarrow \text{displacement} \\ \frac{\partial u}{\partial t}(x,0) = g(x) \rightarrow \text{velocity.} \end{cases}$$

5. Physical derivation: $d = 1$



Let $y_n(t) = u(x_n, t)$, ρ = density > 0, $x_{n+1} - x_n = h$

τ = the tension magnitude > 0.

Use Newton's law: $\tau\left(u_x(x+h,t) - u_x(x,t)\right) = \int_x^{x+h} \rho\, u_{tt}(s,t)\, ds$

Dividing by h and let $h \to 0^+$, gives $u_{tt} = \frac{\tau}{\rho} u_{xx}$ where $c^2 = \frac{\tau}{\rho}$.

Def I.1.2. Consider a linear second-order operator in two variables

$$Lu := a u_{xx} + 2b u_{xy} + c u_{yy} + (\text{lower-order terms})$$

The discriminant of L is: $b^2 - ac$

(1) L is hyperbolic iff $b^2 - ac > 0$

(2) L is parabolic iff $b^2 - ac = 0$.

(3) L is elliptic iff $b^2 - ac < 0$

Def I.1.3 A curve $\gamma(s) = (x(s), t(s))$ is a characteristic (特征线) of $a u_{xx} + 2b u_{xt} + c u_{tt} + \dots = 0$

If it's tangent directions (dx, dt) satisfy $a(dt)^2 - 2b\, dx \cdot dt + c(dx)^2 = 0$

Rmk. the characteristics of the wave equation are: $x - ct = \text{constant}$, $x + ct = \text{constant}$ ($d = 1$)

Let $\xi = x - ct$, $\eta = x + ct$, then $u_{tt} - c^2 u_{xx} = -4c^2 u_{\xi\eta} = 0$

$\Rightarrow u_{\xi\eta} = 0 \Rightarrow u = F(\xi) + G(\eta)$

2. The wave equation in one dimension (Easy condition)

From the previous discussion, we immediately have:

Thm I.2.1. Every C^2-solution of $u_{tt} - c^2 u_{xx} = 0$, $x \in \mathbb{R}$, $t \in \mathbb{R}$ has the form

$u(x,t) = F(x-ct) + G(x+ct)$, F, G be C^2-functions. Converse is also true.

When we are given the initial conditions of the equation, we have the explicit formula:

Thm I.2.2. (d'Alombert's formula) Let $f \in C^2(\mathbb{R})$ and $g \in C^1(\mathbb{R})$. The unique solution of the Cauchy problem is

$$u(x,t) = \frac{1}{2}[f(x+ct) + f(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} g(s)\,ds$$

proof. By (2.1), we have $u(x,t) = F(x-ct) + G(x+ct)$

By the initial conditions, we have $F(x) + G(x) = f(x)$, $-cF'(x) + cG'(x) = g(x)$

Hence $F' + G' = f'$, $-F' + G' = \frac{1}{c}g \Rightarrow F' = \frac{1}{2}(f' - \frac{g}{c})$, $G' = \frac{1}{2}(f' + \frac{g}{c})$

Integrating from x_0 to x, $F(x) = \frac{1}{2}f(x) - \frac{1}{2c}\int_{x_0}^{x} g(t)\,dt + C_1$, $G(x) = \frac{1}{2}f(x) + \frac{1}{2c}\int_{x_0}^{x} g(t)\,dt + C_2$

where $C_1 + C_2 = 0$ ($F + G = f$). hence $u(x,t) = F(x-ct) + G(x+ct)$ = the desired formula □.

Some definitions from (2.2)

Def I.2.3. $D(x_0,t_0) = [x_0 - ct_0, x_0 + ct_0]$ is called the domain of dependence of (x_0,t_0).

Def I.2.4. $I(x_0,0) = \{(x,t) \mid t \geq 0, |x - x_0| \leq ct\}$. is called the domain of influence of $(x_0,0)$

Def I.2.5. $\Gamma_-(x_0,t_0) = \{(x,t) \mid 0 \leq t \leq t_0, |x - x_0| \leq c(t_0 - t)\}$ is called the backward wave cone.

3. Energy, Uniqueness and Well-Posedness

Def I.3.1. For a solution u of the wave equation on $\mathbb{R}^d$ (have compact support in x for every fixed t or decay sufficiently fast at spatial infinity), define the energy:

$$E(t) = \frac{1}{2}\int_{\mathbb{R}^d} \left(|u_t(x,t)|^2 + c^2|\nabla u(x,t)|^2\right) dx \quad \text{, where } |\nabla u|^2 = \sum_{j=1}^{d} |u_{x_j}|^2$$

and

The first term is the kinetic energy, the second is the potential energy (why ?)

Thm I.3.2. If u solves $u_{tt} - c^2\Delta u = 0$ on $\mathbb{R}^d \times [0,T]$ and has compact supp for every fix t.

Then $E(t) = E(0)$. Moreover, $E(0) = \frac{1}{2}\int_{\mathbb{R}^d} (|g|^2 + c^2|\nabla f|^2)$

proof: $\frac{dE}{dt} = \int_{\mathbb{R}^d} (u_t u_{tt} + c^2 \nabla u \cdot \nabla u_t)\,dx$

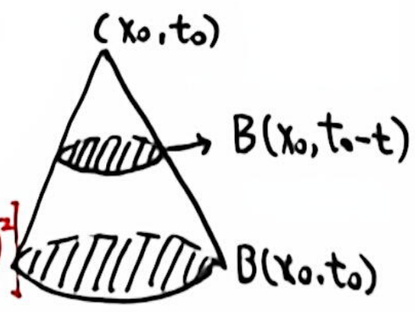
Recall that Green formula: $\iiint_{\partial\Omega} (\nabla u \cdot \nabla v + v \cdot \Delta u) = \oint_{\partial\Omega} v \cdot \frac{\partial u}{\partial \tau}\,ds = 0$!!!

and u has compact supp, we have $\frac{dE}{dt} = \int_{\mathbb{R}^d} (u_t \cdot u_{tt} - c^2 u_t \Delta u)\,dx = 0$

Hence, $E(t) = E(0) = \frac{1}{2}\int_{\mathbb{R}^d} (|g|^2 + c^2|\nabla f|^2)$. □.

Rmk. From the proof we immediately have: $\Omega \subseteq \mathbb{R}^d$ with homogeneous boundary: if $u = 0$ on $\partial\Omega$ (Dirichlet) or $\frac{\partial u}{\partial \tau} = 0$ on $\partial\Omega$ (Neumann), then $\frac{1}{2}\int_\Omega (|u_t|^2 + c^2|\nabla u|^2)\,dx$ is constant. □

Coro. I.3.2. If $u \equiv u_t \equiv 0$ on $B(x_0,t_0) \times \{t \equiv 0\}$. Then $u \equiv 0$ within the cone $\Gamma_-(x_0,t_0)$.



proof (sketch): Define $e(t) = \frac{1}{2}\int_{B(x_0,t_0-t)} (u_t^2 + |\nabla u|^2)\,dx$ $0 \leq t \leq t_0$

Then $\dot{e}(t) = \int_{\partial B} \left(\frac{\partial u}{\partial \nu} u_t - \frac{1}{2}u_t^2 - \frac{1}{2}|\nabla u|^2\right) dS$ where $\left|\frac{\partial u}{\partial \nu} \cdot u_t\right| \leq |u_t||\nabla u| \leq \frac{1}{2}[u_t^2 + (\nabla u)^2]$

$\Rightarrow \dot{e}(t) \leq 0 \Rightarrow e(t) = 0$ □

Thm I.3.3. The Cauchy problem for the wave equation has at most one solution $u \in C^2(\mathbb{R}^d \times [0,T])$ s.t. $u(\cdot,t)$ has compact support in x for each t. (From Rmk, the same holds for...)

proof. Let u, v be two solutions. Let $w = u - v$, then $w(x,0) = w_t(x,0) = 0$

Hence $E_w(t) = E_w(0) = 0 \Rightarrow w_t = 0$, $\nabla w = 0 \Rightarrow w$ is constant. By $w(x,0) = 0 \Rightarrow w \equiv 0$ □.

Thm I.3.4. (Continuous dependence) With u in (3.3), $\frac{1}{2}\int_{\mathbb{R}^d}(|u_t|^2 + |\nabla u|^2 c^2)\,dx = \frac{1}{2}\int_{\mathbb{R}^d}(|g|^2 + c^2|\nabla f|^2)\,dx$, so the map $(f,g) \mapsto (u(\cdot,t), u_t(\cdot,t))$ is an isometry of the natural energy space.

~~4. Fourier transform.~~

~~Recall that for some "nice" functions, we have $\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-2\pi i x\xi}\,dx$, $f(x) = \int_{\mathbb{R}^d} \hat{f}(\xi) e^{2\pi i x\xi}\,d\xi$~~

~~Such functions can be Schwarz functions ($\sup_{x\in\mathbb{R}^d} \left|\frac{\partial f^l}{\partial x}\right| < \infty$ for any l)~~

~~Thm I.4.1. Let $f, g \in S(\mathbb{R}^d)$, then $u(x,t) = \int_{\mathbb{R}^d} \left[\hat{f}(\xi)\cos(2\pi c|\xi|t) + \hat{g}(\xi)\frac{\sin(2\pi c|\xi|t)}{2\pi c|\xi|}\right] e^{2\pi i x\xi}\,d\xi$~~

~~solves the Cauchy problem. And $u(\cdot,t) \in S(\mathbb{R}^d)$, $\forall t > 0$.~~

4. Spherical means.

DefI.4.1. Let $x \in \mathbb{R}^n$, $t > 0$, $r > 0$. Define $U(x;r,t) := \fint_{\partial B(x,r)} u(y,t)\,dS(y)$ $\partial B(x,r)$ is the sphere.

Similarly, $\begin{cases} G(x;r) := \fint_{\partial B(x,r)} g(y)\,dS(y) \\ H(x;r) := \fint_{\partial B(x,r)} h(y)\,dS(y). \end{cases}$

Lemma I.4.2. (Euler-Poisson-Darboux equation) Fix $x \in \mathbb{R}^n$. and u satisfies the Cauchy problem,

Then $\begin{cases} U \text{ in (4.1) satisfies} \\ u \in C^m(\overline{\mathbb{R}^+} \times [0,+\infty)) \end{cases}$ $\begin{cases} U_{tt} - U_{rr} - \frac{n-1}{r}U_r = 0 & \text{in } \mathbb{R}_+ \times (0,+\infty) \\ U = G,\ U_t = H & \text{on } \mathbb{R}_+ \times \{t=0\} \end{cases}$

proof: $\forall r > 0$, we have $U_r(x;r,t) = \frac{\partial}{\partial r}\left(\frac{1}{4\pi r^2}\int_{\partial B(x,r)} u(y,t)\,dS(y)\right)$

$= \frac{\partial}{\partial r}\left(\fint_{\partial B(0,1)} u(x+rz)\,dS(z)\right) = \fint_{\partial B(0,1)} Du(x+rz)\cdot z\,dS(z)$

$= \fint_{\partial B(x,r)} Du(y)\cdot\frac{y-x}{r}\,dS(y) = \fint_{\partial B(x,r)} \frac{\partial u}{\partial \nu}\,dS(y) \xrightarrow{\text{Green's formula}} \frac{r}{n}\fint_{B(x,r)} \Delta u(y,t)\,dy$

Hence $U_r(x;r,t) = \frac{r}{n}\fint_{B(x,r)} \Delta u(y,t)\,dy$ (*)

differientiate it, $U_{rr} = \frac{1}{n}\fint_{B(x,r)} \Delta u\,dy + \fint_{\partial B(x,r)} \Delta u\,dS - \fint_{B(x,r)} \Delta u\,dy$

$\Rightarrow \lim_{r\to 0^+} U_{rr}(x;r,t) = \frac{1}{n}\Delta u(x,t)$. So it's easy to check that $U \in C^m(\overline{\mathbb{R}_+} \times [0,+\infty))$

By the wave equation: (*) $\Rightarrow$ $U_r = \frac{r}{n}\fint_{B(x,r)} U_{tt}\,dy = \frac{1}{n\alpha(n)}\frac{1}{r^{n-1}}\fint_{B(x,r)} U_{tt}\,dy$

So $(r^{n-1}U_r)_r = \frac{1}{n\alpha(n)}\int_{\partial B(x,r)} U_{tt}\,dS = r^{n-1}\fint_{\partial B(x,r)} U_{tt}\,dS = r^{n-1}U_{tt}$.

Namely, $U_{rr} + \frac{n-1}{r}U_r = U_{tt}$ □. (Amazing)

By this lemma, we can give an explicit formula for wave equations.

Case 1. $(2+n)$. $\underline{n=2k+1}$.

Thm I§2.3. Assume that $2+n$, $n\geq 3$. Suppose that $g\in C^{m+1}(\mathbb{R}^n)$, $h\in C^{m}(\mathbb{R}^n)$, for $m=\frac{n+1}{2}$.

Then the Cauchy problem $\begin{cases} u_{tt}-\Delta u=0 \\ u(x,0)=g(x) \\ u_t(x,0)=h(x) \end{cases}$ has a $\underline{C^2\text{-solution}}$ u

defined by $u(x,t)=\frac{1}{(n-2)!!}\left[\left(\frac{\partial}{\partial t}\right)\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-3}{2}}\left(t^{n-2}\int_{\partial B(x,t)} g\,dS\right)+\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-3}{2}}\left(t^{n-2}\int_{\partial B(x,t)} h\,dS\right)\right]$

proof: We need some notations to simplify our writings

Lemma: $\phi:\mathbb{R}\to\mathbb{R}$ be C^{k+1}. Then (i) $\left(\frac{d^2}{dr^2}\right)\left(\frac{1}{r}\frac{d}{dr}\right)^{k-1}\left(r^{2k-1}\phi(r)\right)=\left(\frac{1}{r}\frac{d}{dr}\right)^{k}\left(r^{2k}\frac{d\phi}{dr}(r)\right)$

(ii) $\left(\frac{1}{r}\frac{d}{dr}\right)^{k-1}\left(r^{2k-1}\phi(r)\right)=\sum_{j=1}^{k-1}\beta_j^k r^{j+1}\frac{d^j\phi}{dr^j}(r)$ where β_j^k are constan

(iii) $\beta_0^k=(2k-1)!!$

We skip the proof.

Now we write: $\begin{cases} \tilde{u}(r,t):=\left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1}\left(r^{2k-1}u(x;r,t)\right) \\ \tilde{G}(r):=\left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1}\left(r^{2k-1}G(x;r)\right) \\ \tilde{H}(r):=\left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1}\left(r^{2k-1}H(x;r)\right) \end{cases}=$ Then $\begin{cases} \tilde{u}(r,0)=\tilde{G} \\ \tilde{u}_t(r,0)=\tilde{H} \end{cases}$

Step 1. Find solution

We claim that $\tilde{u}$ solves the 1-dim wave equation, i.e. $\begin{cases} \tilde{u}_{tt}-\tilde{u}_{rr}=0 \\ \tilde{u}=\tilde{G},\ \tilde{u}_t=\tilde{H} \quad \text{when } t=0 \\ \tilde{u}=0 \quad \text{when } r=0 \text{ \& } t=0 \end{cases}$

Indeed, $\tilde{u}_{rr}=\left(\frac{\partial^2}{\partial r^2}\right)\left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1}\left(r^{2k-1}u\right)\overset{\text{lemma}}{=\!=}\left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k}\left(r^{2k}u_r\right)$

$=\left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1}\left(r^{2k-1}u_{rr}+2kr^{2k-2}u_r\right)=\left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1}\left[r^{2k-1}\left(u_{rr}+\frac{n-1}{r}u_r\right)\right]$

$=\left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1}\left(r^{2k-1}u_{tt}\right)=\tilde{u}_{tt}$. $\square$.

Now recall the d'Alembert's formula: $\tilde{u}(r,t)=\frac{1}{2}\left[\tilde{G}(r+t)-\tilde{G}(r+t)\right]+\frac{1}{2}\int_{t-r}^{t+r}\tilde{H}(y)\,dy$

And by lemma (ii) & (iii). $\tilde{u}(r,t)=\sum_{j=0}^{k-1}\beta_j^k r^{j+1}\frac{\partial^j}{\partial r^j}u(x;r,t)$

$$\lim_{r\to 0^+}\frac{\tilde{u}(r,t)}{\beta_0^k r}=\lim_{r\to 0^+}u(x;r,t)=u(x,t)$$

$$\text{LHS}=\frac{1}{\beta_0^k}\lim_{r\to 0^+}\left(\frac{\tilde{G}(r+t)-\tilde{G}(t-r)}{2r}+\frac{1}{2r}\int_{t-r}^{t+r}\tilde{H}(y)\,dy\right)=\frac{1}{\beta_0^k}\left[\tilde{G}'(t)+\tilde{H}(t)\right]$$ where we derive the desired formula.

Step 2. Check the solution:

W.L.O.G let $g\equiv 0$, then $u(x,t)=\frac{1}{\gamma_n}\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-3}{2}}\left(t^{n-2}H(x;t)\right)$, $\gamma_n=\frac{1}{(n-2)!!}$.

Then $u_{tt}=\frac{1}{\gamma_n}\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-1}{2}}\left(t^{n-1}H_t\right)$ (by lemma)

where $H_t=\frac{\partial}{\partial t}\left(\int_{\partial B(x,t)}h(y)dS(y)\right)\overset{\text{Similar in (4.2)}}{=\!=}\frac{t}{n}\int_{B(x,t)}\Delta h\,dy$

Therefore, $u_{tt}=\frac{1}{\gamma_n\cdot n\alpha(n)}\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-3}{2}}\left(\frac{1}{t}\int_{\partial B(x,t)}\Delta h\,dS\right)$

$$\Rightarrow \Delta u = \Delta\left(\frac{1}{r_n n\alpha(n)}\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-3}{2}}\left(\frac{1}{t}\int_{\partial B(x,t)} h(y)\,dS(y)\right)\right) = \frac{1}{r_n\cdot n\alpha(n)}\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-3}{2}}\left(\frac{1}{t}\int_{\partial B(x,t)}\Delta h\,dS\right)$$

$$= u_{tt} \quad \square$$

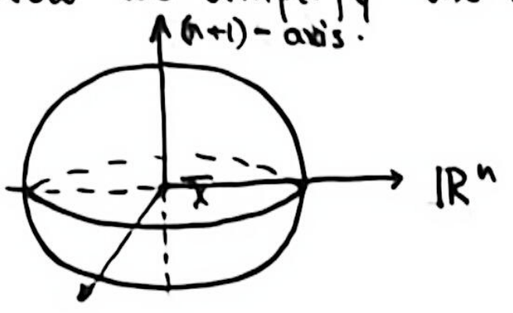
Case 2. $(2|n)$. The trick is to write : $\overline{u}(x_1,\dots,x_{n+1},t) := u(x_1,\dots x_n,t)$

Then it solves the following $(n+1)$-dim Cauchy problem:

$$\begin{cases} \overline{u}_{tt} - \Delta\overline{u} = 0 \\ \overline{u} = \overline{g},\ \overline{u_t} = \overline{h} \quad \text{on } \mathbb{R}^{n+1}\times\{t=0\} \end{cases}$$

By previous theory, we have $u(x,t) = \frac{1}{r_{n+1}}\left[\frac{\partial}{\partial t}\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-2}{2}}\left(t^{n-1}\int_{\partial\overline{B}(\overline{x},t)}\overline{g}\,d\overline{s}\right) + \left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-2}{2}}\left(t^{n-1}\int_{\partial\overline{B}(\overline{x},t)}\overline{h}\,d\overline{s}\right)\right]$

Now we simplify the result: $\overline{x} = (x,0)$ where $x\in\mathbb{R}^n$



By Fubini's thm, we have

$$\int_{\partial\overline{B}(\overline{x},t)}\overline{g}\,d\overline{s} = \frac{2}{(1+n)\alpha(n+1)\,t^n}\int_{B(x,t)} g(y)\frac{1}{\sqrt{t^2-|y-x|^2}}\,dy =$$

$$= \frac{2t\alpha(n)}{(n+1)\alpha(n+1)}\int_{B(x,t)}\frac{g(y)}{\sqrt{t^2-|y-x|^2}}\,dy. \quad \left(\text{Similarly for } \int_{\partial\overline{B}(\overline{x},t)}\overline{h}\,d\overline{s}\right).$$

$\left(\text{// } \sqrt{1+(\nabla r)^2} \text{ where } r(y) = \sqrt{t^2-(y-x)^2},\ y\in B(x,t)\right)$

Therefore. $u(x,t) = \frac{1}{r_n}\left[\frac{\partial}{\partial t}\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-2}{2}}\left(t^n\int_{B(x,t)}\frac{g(y)\,dy}{\sqrt{t^2-(y-x)^2}}\right) + \left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-2}{2}}\left(t^n\int_{B(x,t)}\frac{h(y)}{\sqrt{t^2-(y-x)^2}}\,dy\right)\right]$ (**)

Now we get the following theorem:

Thm I4.4 $(2|n, n\geq 2)$ Suppose $g\in C^{m+1}(\mathbb{R}^n)$, $h\in C^m(\mathbb{R}^n)$ for $m=\frac{n+2}{2}$. Then u defined by (**) is the unique solution to the Cauchy problem $\square$.

Rmk. Huygen's principle. (惠更斯) (1) If $2\nmid n$, $n\geq 3$; the data g and h at x affect u only on the boundary $\{(y,t)\mid t>0;\ |x-y|=t\}$

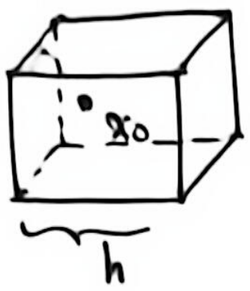
(2) If $2|n$, $n\geq 2$. the data g and h at x affect u in $\{(y,t)\mid t>0,\ |x-y|\leq t\}$

Ⅱ. Heat equation

1. Def & Fundamental solutions.

Def Ⅱ.1.1. $u_t - \Delta u = 0$ (*) is called the heat equation.

Rmk 1. Physical derivation: Let $u(x,t)$ denote the temperature at $x \in \mathbb{R}^n$ when time is t.



Consider the heat energy in S cube with length h

$H(t) = 6\int_S u(x,t)\,dx$. then the heat flow into S is $\frac{\partial H}{\partial t} = 6\int_S \frac{\partial u}{\partial t}\,dx$.

$\approx 6h^n \frac{\partial u}{\partial t}$

By New's cooling law,

$\frac{\partial H}{\partial t} = \bullet\, k h^{n-1} \nabla u \Rightarrow \frac{\partial u}{\partial t} = \frac{6}{k}\frac{\nabla u}{h}$ let $h \to 0^+$, we have $\frac{\partial u}{\partial t} = \Delta u \cdot \frac{6}{k}$ □.

2. If u is the solution, then so does $u(\lambda x, \lambda^2 t)$ for $\lambda \in \mathbb{R}$ ($v(x,t) = u(\lambda x, \lambda^2 t)$

$v_t = \lambda^2 u_t$, $\Delta v = \lambda^2 \Delta u$)

3. If we find some special solutions like $u(x,t) = \frac{1}{t^\alpha} v\left(\frac{x}{t^\beta}\right)$ $(x \in \mathbb{R}^n, t>0)$.

and $u(x,t) = \lambda^\alpha u(\lambda^\beta t, \lambda t)$ for $\lambda>0$, $t>0$, $x \in \mathbb{R}^n$.

By some computations, we have $u(x,t) = \frac{b}{t^{\frac{n}{2}}} e^{-\frac{|x|^2}{4t}}$ solves the equation.

Def Ⅱ.1.2. $\Phi(x,t) = \begin{cases} \frac{1}{(4\pi t)^{\frac{n}{2}}} e^{-\frac{|x|^2}{4t}} & (t>0,\ x \in \mathbb{R}^n) \\ 0 & (t<0,\ x \in \mathbb{R}^n) \end{cases}$ is called the foundamental solution of (*)

Lemma Ⅱ.1.3. ($\Phi(\cdot,t)$ is a good kernel $\forall t>0$) $\forall t>0$, $\int_{\mathbb{R}^d} \Phi(x,t)\,dx = 1$.

pf: $\int_{\mathbb{R}^n} \Phi(x,t)\,dx = \frac{1}{(4\pi t)^{n/2}} \int_{\mathbb{R}^n} \exp(-|x|^2/4t)\,dx = \frac{1}{\pi^{\frac{n}{2}}} \prod_{k\geq 1} \int_{\mathbb{R}} e^{-x_i^2}\,dx_i = 1$ □.

Thm Ⅱ.1.4. Assume $g \in C(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$, and $u = \Phi * g$. Then

(i) $u \in C^\infty(\mathbb{R}^n \times (0,+\infty))$

(ii) $u_t - \Delta u = 0$ on $\mathbb{R}^n \times (0,+\infty)$

(iii) $u = g$ on $\mathbb{R}^n \times \{t=0\}$.

proof: (i) is immediate by the property of convolution.

(ii) $u_t - \Delta u = \int_{\mathbb{R}^n} [(\Phi_t - \Delta\Phi)(x-y,t)] \cdot g(y)\,dy = 0$.

(iii). ~~$\forall x \in \mathbb{R}^n, \varepsilon>0, \exists \delta>0$ s.t. $|g(y)-g(x)|<\varepsilon$. $\forall |y-x|<\delta$.~~

$\Phi(\cdot,t)$ is a good kernel □.

2. Mean-value formula

Def II.2.1. Let $U \subseteq \mathbb{R}^n$ open and bounded, $T>0$; (i) $U_T := U \times (0,T]$

(ii) $\Gamma_T := \overline{U_T} - U_T$.

(iii) $E(x,t;r) := \{(y,s) \in \mathbb{R}^{n+1} \mid s \le t,\ \Phi(x-y,t-s) \ge \frac{1}{r^n}\}$. is called the heat ball.

where the boundary of $E(x,t;r)$ is a level set of Φ.

Thm II.2.2. Let $u \in C_1^2(U_T)$ solve the heat equation. Then $u(x,t) = \frac{1}{4r^n} \iint_{E(x,t;r)} u(y,s) \frac{|x-y|^2}{(t-s)^2} dy ds$

$\forall E(x,t;r) \subseteq U_T$ (It's similar to Mean-value formula in harmonic functions).

proof: W.L.O.G let $x=0$, $t=0$ and $E(r) = E(0,0;r)$.

Let $\phi(r) := \frac{1}{r^n} \iint_{E(r)} u(y,s) \frac{|y|^2}{s^2} dy ds = \iint_{E(1)} u(ry, r^2 s) \frac{|y|^2}{s^2} dy ds$

Then $\phi'(r) = \iint_{E(1)} \left(\sum_{i=1}^n u_{y_i} \cdot y_i \frac{|y|^2}{s^2} + 2r u_s \frac{|y|^2}{s^2} \right) dy ds$

$= \frac{1}{r^{n+1}} \iint_{E(r)} \left(\sum_{i=1}^n u_{y_i} y_i \frac{|y|^2}{s^2} + 2 u_s \frac{|y|^2}{s^2} \right) dy ds := A + B$.

Define $\psi(y,s) = -\frac{n}{2} \log(-4\pi s) + \frac{|y|^2}{4s} + n \log r$.; $\psi = 0$ on $\partial E_r - \{(0,0)\}$.

And $\Phi(y,-s) = r^{-n}$ on ∂E_r, we have:

$$B = \frac{1}{r^{n+1}} \iint_{E(r)} 4 u_s \sum_{i=1}^n y_i \psi_{y_i} \, dy ds = -\frac{1}{r^{n+1}} \iint_{E(r)} 4n u_s \psi + 4 \sum_{i=1}^n u_{s y_i} y_i \psi \, ds ds$$

Integrating by parts: $\left(-\frac{n}{2s} - \frac{|y|^2}{4s^2}\right)$.

$$B = \frac{1}{r^{n+1}} \iint_{E(r)} -4n u_s \psi + 4 \sum_{i=1}^n u_{y_i} \cdot y_i \cdot \psi_s \, dy ds$$

$$= \frac{1}{r^{n+1}} \iint_{E(r)} -4n u_s \psi - \frac{2n}{s} \sum_{i=1}^n u_{y_i} y_i \, dy ds - A$$

$u_s = \Delta u$.

Therefore $\phi(r) = A + B = (\text{Use Green formula}) = \sum_{i=1}^n \frac{1}{r^{n+1}} \iint_{E(r)} \left(4n u_{y_i} \psi_{y_i} - \frac{2n}{s} u_{y_i} \cdot y_i\right) dy ds$

$= 0$

$\Rightarrow \phi(r) = \lim_{t \to 0} \phi(t) = u(0,0) \lim_{t \to 0} \frac{1}{t^n} \iint_{E(t)} \frac{|y|^2}{|s|^2} dy ds = 4 u(0,0)$ □.

By this formula, we can prove some properties of solutions.

Thm II.2.3. (Maximum principle) $u \in C_1^2(U_T) \cap C(\overline{U_T})$ solves the heat equation in U_T

(i) $\max_{\overline{U_T}} u = \max_{\Gamma(T)} u$; (ii) If U is connected and $\exists (x_0,t_0) \in U_T$ s.t. $u(x_0,t_0) = \max_{\overline{U_T}} u$.

then u is constant in $\overline{U_{t_0}}$.

proof: (i) Assume that $\exists (x_0,t_0) \in U_T$ s.t. $u(x_0,t_0) = M$

Then $\exists$ small $r>0$, s.t. $M = u(x_0,t_0) = \frac{1}{4r^n} \iint_{E(x_0,t_0;r)} u(y,s) \frac{|x_0-y_0|^2}{(t_0-s)^2} dy ds \le M$.

$\Rightarrow u \equiv M$ in $E(x_0,t_0;r)$

$\forall (y_0,s_0) \in U_T$. Let L be the line connecting (x_0,t_0) & (y_0,s_0), $(s_0<t_0)$

let $r_0 = \min\{s \ge s_0 \mid u(x,t) = M,\ s \le t \le t_0\}$. It's easy to check $r_0 = s_0 \Rightarrow u \equiv M$ on L □

8.

Thm II.2.4. It's easy to turn out the uniqueness for the Cauchy problem (for bounded domain) of the heat equation.

Now we estimate the solutions. Suppose $u \in C_1^2(\mathbb{R}^n)(0,T]) \cap C(\mathbb{R}^n \times (0,T])$ solves

$$\begin{cases} u_t - \Delta u = 0 & \text{in } \mathbb{R}^n \times (0,T) \\ u = g & \text{on } \mathbb{R}^n \times \{t=0\}. \end{cases}$$

and satisfies $u(x,t) \le A e^{a|x|^2}$ $(0 \le t \le T)$. $a>0$

Then $\sup_{\mathbb{R}^n \times [0,T]} u = \sup_{\mathbb{R}^n} g$

proof: Case 1: $4aT < 1$. $4a(T+\varepsilon) < 1$ for some $\varepsilon > 0$ Fix $y \in \mathbb{R}^n$, $\mu > 0$.

define $v(x,t) := u(x,t) - \dfrac{\mu}{(T+\varepsilon-t)^{n/2}} e^{\frac{|x-y|^2}{4(T+\varepsilon-t)}}$ $(t>0)$

Then $v_t - \Delta v = 0$ in $\mathbb{R}^n \times (0,T]$. $\forall r>0$, set $U := B^\circ(y,r)$, $U_T = B^\circ(y,r) \times (0,T]$

we have $\max_{\overline{U_T}} v = \max_{\Gamma(T)} v$. $(*)$

Now if $x \in \mathbb{R}^n$, $v(x,0) = u(x,0) - \dfrac{\mu}{(T+\varepsilon)^{n/2}} e^{\frac{|x-y|^2}{4(T+\varepsilon)}} \le u(x,0) = g(x)$

If $|x-y| = r$, $0 \le t \le T$, we have $v(x,t) = u(x,t) - \dfrac{\mu}{(T+\varepsilon-t)^{\frac{n}{2}}} e^{\frac{r^2}{4(T+\varepsilon-t)}}$

$$\le A e^{\underbrace{a|x|^2}_{\| \\ (|y|+r)^2}} - \frac{\mu}{(T+\varepsilon)^{\frac{n}{2}}} e^{\frac{r^2}{4(T+\varepsilon)}}$$

Since $4a(T+\varepsilon) < 1$, we conclude that

$$v(x,t) \le A e^{(|y|+r)^2} - \mu \left(4(a+\gamma)\right)^{n/2} e^{(a+\gamma) r^2} \le \sup_{\mathbb{R}^n} g \quad \text{(for } r \text{ large enough)}$$

By $(*)$, we have $v(y,t) \le \sup_{\mathbb{R}^n} g$ $\forall\, y \in \mathbb{R}^n$, $0 \le t \le T$ Let $\mu \to 0 \Longrightarrow \sup_{\mathbb{R}^n \times [0,T]} u \le \sup_{\mathbb{R}^n} g$ □

Case 2: $4aT \ge 1$, divide $[0,T]$ into smaller intervals □

Rmk. 1. From this result, we can get the uniqueness of the Cauchy problem on $\mathbb{R}^n$

2. The striking result is that $u \in C_1^2(U_T) \Rightarrow u \in C^\infty(U_T)$ has nice regularity. we skip its proof.

3. We can define the energy: $e(t) = \int_u u^2(x,t)\,dx$ $(0 \le t \le T)$.

And $\frac{d}{dt} e(t) = 2\int_u u \cdot u_t \, dx \overset{\text{Equation}}{=\!=} 2\int_u u \cdot \Delta u \, dx \overset{\text{Green}}{=\!=} -2\left(\int_u |\nabla u|^2 dx + \int_{\partial u} \frac{\partial u}{\partial v} \cdot u \, dS\right)$

If $u = \mathbb{R}^n$ or $u \equiv 0$ on ∂u. Then $\frac{d}{dt} e(t) = -2\int_u |\nabla u|^2 dx \le 0$

4. From this we can get a new proof of the Uniquess of the Cauchy probelm for $(x,t) \in U_T$ □.